How Big Is the Bootstrap World?

- The number of distinct bootstrap samples of size n is $\binom{2n-1}{n}$:

$n = 3$	$n = 10$	$n = 20$
10	92,378	$\approx 6.9 \times 10^{10}$

- Exact enumeration is hopeless beyond tiny n – so we approximate the ideal bootstrap by drawing B resamples at random: **Monte Carlo inside the bootstrap world**.

Two separate layers of approximation

$$\underbrace{\mathrm{se}_F(\hat{\theta})}_{\text{truth}} \overset{\text{statistical error}}{\approx} \underbrace{\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*)}_{\text{ideal bootstrap}} \overset{\text{Monte Carlo error}}{\approx} \underbrace{\widehat{\mathrm{se}}_B}_{\text{what we compute}}$$

Increasing B removes only the second error. The first is fixed by the data (it shrinks only with n).

Bootstrap Standard Error and Bias

Bootstrap estimates from B replications

With $\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^*_b$,

$$\widehat{\text{se}}_B = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}^*_b - \bar{\theta}^*\right)^2}, \qquad \widehat{\text{bias}}_B = \bar{\theta}^* - \hat{\theta}.$$

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients — small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

$\Rightarrow se(\bar{x}) = \frac{s}{\sqrt{n}} \quad \Rightarrow \bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

$V(\bar{x}) = \frac{\sigma^2}{n}$

$se(\bar{x}) = \frac{\sigma}{\sqrt{n}} \rightarrow \frac{s}{\sqrt{n}}$

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

Bootstrap ($B = 4999$)

$\widehat{\text{se}}_B = 0.571 \quad (\approx \sqrt{19/20} \times 0.585)$

$\widehat{\text{bias}}_B = 0.002 \quad$ (negligible, as theory says)

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

Bootstrap ($B = 4999$)

$\widehat{\text{se}}_B = 0.571 \quad (\approx \sqrt{19/20} \times 0.585)$

$\widehat{\text{bias}}_B = 0.002 \quad$ (negligible, as theory says)

- For the mean, the bootstrap simply reproduces theory – a useful **sanity check** of your code.

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

Bootstrap ($B = 4999$)

$\widehat{\text{se}}_B = 0.571 \quad (\approx \sqrt{19/20} \times 0.585)$

$\widehat{\text{bias}}_B = 0.002 \quad$ (negligible, as theory says)

- For the mean, the bootstrap simply reproduces theory – a useful **sanity check** of your code.
- The payoff comes when you switch $s(\cdot)$ to the *median* or a 10% *trimmed mean*: the same five lines of code give a standard error that classical theory cannot.

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

Bootstrap ($B = 4999$)

$\widehat{\text{se}}_B = 0.571 \quad (\approx \sqrt{19/20} \times 0.585)$

$\widehat{\text{bias}}_B = 0.002 \quad$ (negligible, as theory says)

- For the mean, the bootstrap simply reproduces theory – a useful **sanity check** of your code.
- The payoff comes when you switch $s(\cdot)$ to the *median* or a 10% *trimmed mean*: the same five lines of code give a standard error that classical theory cannot.

We will run this example (and the median version) in the lab.

Why Does the Bootstrap Work? (Sketch)



- The bootstrap approximates the law of a *root*, e.g. $R = \sqrt{n}(\hat{\theta} - \theta)$ under F, by the law of $R^* = \sqrt{n}(\hat{\theta}^* - \hat{\theta})$ under $\hat{F}_n$.

Why Does the Bootstrap Work? (Sketch)

- The bootstrap approximates the law of a *root*, e.g. $R = \sqrt{n}(\hat{\theta} - \theta)$ under F, by the law of $R^* = \sqrt{n}(\hat{\theta}^* - \hat{\theta})$ under $\hat{F}_n$.

Theorem (Bootstrap consistency for the mean; Bickel & Freedman, 1981)

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$ with $\mathbb{E}X^2 < \infty$, then almost surely

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_*\left(\sqrt{n}(\bar{X}^* - \bar{X}) \le x\right) - \mathbb{P}_F\left(\sqrt{n}(\bar{X} - \mu) \le x\right) \right| \longrightarrow 0.$$

The result extends to smooth functions of means (delta method), and many other smooth statistics.

Why Does the Bootstrap Work? (Sketch)

- The bootstrap approximates the law of a *root*, e.g. $R = \sqrt{n}(\hat{\theta} - \theta)$ under F, by the law of $R^* = \sqrt{n}(\hat{\theta}^* - \hat{\theta})$ under $\hat{F}_n$.

Theorem (Bootstrap consistency for the mean; Bickel & Freedman, 1981)

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$ with $\mathbb{E}X^2 < \infty$, then almost surely

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_*\big(\sqrt{n}(\bar{X}^* - \bar{X}) \le x\big) - \mathbb{P}_F\big(\sqrt{n}(\bar{X} - \mu) \le x\big) \right| \longrightarrow 0.$$

The result extends to smooth functions of means (delta method), and many other smooth statistics.

- **Intuition:** the law of a good root depends on F only through a few features (variance, skewness, ...), and $\hat{F}_n$ gets those features right as n grows.

Why Does the Bootstrap Work? (Sketch)

- The bootstrap approximates the law of a *root*, e.g. $R = \sqrt{n}(\hat{\theta} - \theta)$ under F, by the law of $R^* = \sqrt{n}(\hat{\theta}^* - \hat{\theta})$ under $\hat{F}_n$.

Theorem (Bootstrap consistency for the mean; Bickel & Freedman, 1981)

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$ with $\mathbb{E}X^2 < \infty$, then almost surely

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_*\big(\sqrt{n}(\bar{X}^* - \bar{X}) \le x\big) - \mathbb{P}_F\big(\sqrt{n}(\bar{X} - \mu) \le x\big) \right| \longrightarrow 0.$$

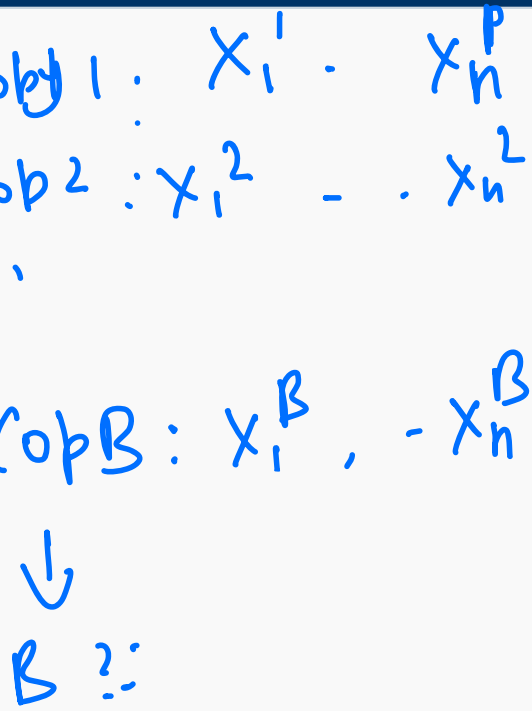
The result extends to smooth functions of means (delta method), and many other smooth statistics.

- **Intuition:** the law of a good root depends on F only through a few features (variance, skewness, ...), and $\hat{F}_n$ gets those features right as n grows.
- The theorem needs **smoothness** and **finite variance** – we will see what happens when they fail (Section 9).

How Many Bootstrap Resamples B?

Rules of thumb (Efron & Tibshirani, 1994)

- Standard error: $B = 200$ is usually plenty.
- Percentile-based confidence intervals: $B \geq 1000$–2000 (tails need more draws).
- p-values: $B = 999$ or 9999 (see Section 7).



How Many Bootstrap Resamples B?

Rules of thumb (Efron & Tibshirani, 1994)

- Standard error: $B = 200$ is usually plenty.
- Percentile-based confidence intervals: $B \geq 1000$–2000 (tails need more draws).
- p-values: $B = 999$ or 9999 (see Section 7).

- Monte Carlo error of anything bootstrap-based shrinks like $1/\sqrt{B}$ – the Module 2 rate.

How Many Bootstrap Resamples B?

Rules of thumb (Efron & Tibshirani, 1994)

- Standard error: $B = 200$ is usually plenty.
- Percentile-based confidence intervals: $B \geq 1000$–2000 (tails need more draws).
- p-values: $B = 999$ or 9999 (see Section 7).

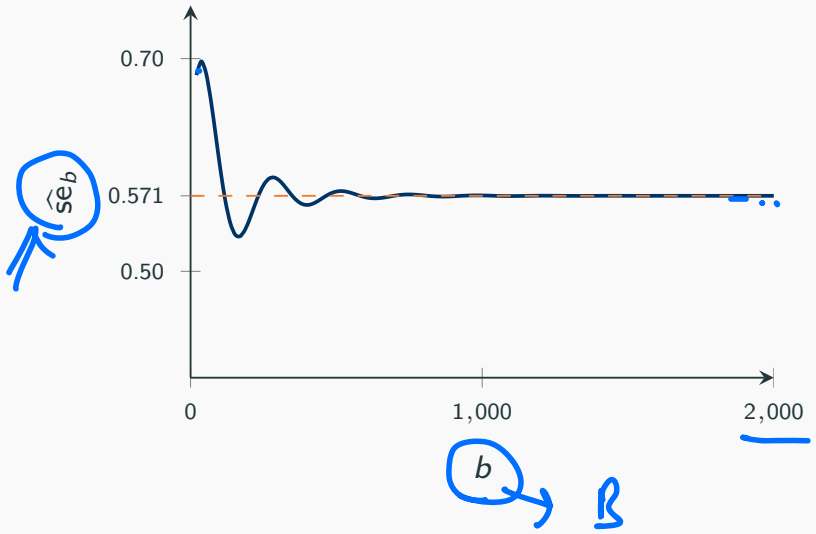
- Monte Carlo error of anything bootstrap-based shrinks like $1/\sqrt{B}$ – the Module 2 rate.
- Use the Module 2 **running-mean trace plot** on $\widehat{\text{se}}_b$ against b: stop once it is flat.

How Many Bootstrap Resamples B?

Rules of thumb (Efron & Tibshirani, 1994)

- Standard error: $B = 200$ is usually plenty.
- Percentile-based confidence intervals: $B \geq 1000$–2000 (tails need more draws).
- p-values: $B = 999$ or 9999 (see Section 7).

- Monte Carlo error of anything bootstrap-based shrinks like $1/\sqrt{B}$ – the Module 2 rate.
- Use the Module 2 **running-mean trace plot** on $\widehat{\text{se}}_b$ against b: stop once it is flat.



trace settles near the ideal value

Outline

1. Motivation: Inference When Formulas Run Out
2. The Bootstrap Principle
3. Parametric vs Nonparametric Bootstrap
   - The Parametric Bootstrap
   - Example: Exponential Lifetimes
   - Bootstrapping Regression Models
   - Dependent Data: The Block Bootstrap
   - Choosing Between Them
4. Bootstrap Confidence Intervals
5. The Jackknife (Brief)
6. Cross-Validation
7. Monte Carlo Hypothesis Testing and Permutation Tests
8. Likelihood-Based Inference via Simulation
9. Simulation vs Asymptotic Inference
10. Unit Summary

The Parametric Bootstrap

- If we trust a parametric model $\{F_\theta\}$, estimate F by $F_{\hat{\theta}}$ instead of $\hat{F}_n$.

The Parametric Bootstrap

- If we trust a parametric model $\{F_\theta\}$, estimate F by $F_{\hat{\theta}}$ instead of $\hat{F}_n$.
- We then simulate **new** data from the fitted model – using the generators of Module 1.

The Parametric Bootstrap

- If we trust a parametric model $\{F_\theta\}$, estimate F by $F_{\hat{\theta}}$ instead of $\hat{F}_n$.
- We then simulate **new** data from the fitted model – using the generators of Module 1.

Algorithm: Parametric Bootstrap

1. Fit the model: $\hat{\theta}$ (typically the MLE) from $\boldsymbol{x}$.
2. For $b = 1, \ldots, B$:
   (a) Simulate $x_1^{*b}, \ldots, x_n^{*b} \overset{\text{iid}}{\sim} F_{\hat{\theta}}$.
   (b) Re-fit the model to $\boldsymbol{x}^{*b}$; record $\hat{\theta}_b^*$ (or any statistic of interest).
3. Use $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ exactly as in the nonparametric case.

The Parametric Bootstrap

- If we trust a parametric model $\{F_\theta\}$, estimate F by $F_{\hat{\theta}}$ instead of $\hat{F}_n$.
- We then simulate **new** data from the fitted model – using the generators of Module 1.

Algorithm: Parametric Bootstrap

1. Fit the model: $\hat{\theta}$ (typically the MLE) from $\boldsymbol{x}$.
2. For $b = 1, \ldots, B$:
   (a) Simulate $x_1^{*b}, \ldots, x_n^{*b} \overset{\text{iid}}{\sim} F_{\hat{\theta}}$.
   (b) Re-fit the model to $\boldsymbol{x}^{*b}$; record $\hat{\theta}_b^*$ (or any statistic of interest).
3. Use $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ exactly as in the nonparametric case.

Key difference

Nonparametric bootstrap: *resamples the observed values*. Parametric bootstrap: *generates new values* from a fitted model – so it can produce values never seen in the data (e.g. more extreme ones).

Example: Exponential Lifetimes (Reliability Engineering)

- Lifetimes of n components are modelled as $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Engineers report the **median lifetime** $m = \ln 2/\lambda$.

$X_1, \ldots, X_n$

Example: Exponential Lifetimes (Reliability Engineering)

- Lifetimes of n components are modelled as $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Engineers report the **median lifetime** $m = \ln 2/\lambda$.
- MLE: $\hat{\lambda} = 1/\bar{x}$, so $\hat{m} = \bar{x} \ln 2$.

Example: Exponential Lifetimes (Reliability Engineering)

- Lifetimes of n components are modelled as $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Engineers report the **median lifetime** $m = \ln 2/\lambda$.
- MLE: $\hat{\lambda} = 1/\bar{x}$, so $\hat{m} = \bar{x} \ln 2$.

Parametric bootstrap

1. Simulate $X_i^* = -\ln(U_i)/\hat{\lambda}, \quad U_i \overset{\text{iid}}{\sim} \text{Unif}(0,1)$ (inverse transform, Module 1).
2. Compute $\hat{m}^* = \bar{X}^* \ln 2$; repeat B times; take the standard deviation.

Example: Exponential Lifetimes (Reliability Engineering)

- Lifetimes of n components are modelled as $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Engineers report the **median lifetime** $m = \ln 2/\lambda$.
- MLE: $\hat{\lambda} = 1/\bar{x}$, so $\hat{m} = \bar{x} \ln 2$.

Parametric bootstrap

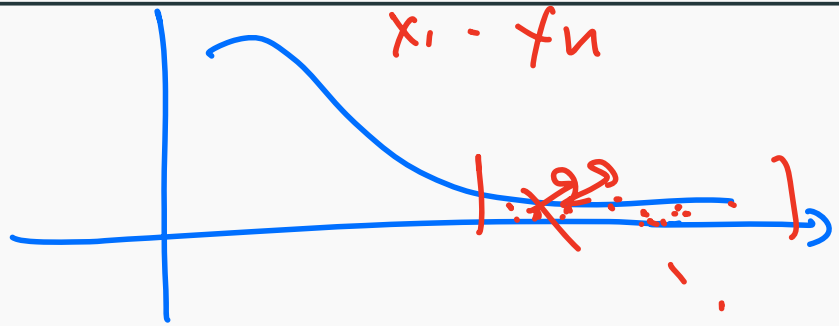
1. Simulate $X_i^* = -\ln(U_i)/\hat{\lambda}$, $U_i \overset{\text{iid}}{\sim} \text{Unif}(0,1)$ (inverse transform, Module 1).
2. Compute $\hat{m}^* = \bar{X}^* \ln 2$; repeat B times; take the standard deviation.

Check against exact theory

Under $F_{\hat{\lambda}}$, $\bar{X}^* \sim \text{Gamma}(n,\ \text{rate}\ n\hat{\lambda})$, so $\text{se}(\hat{m}^*) = \bar{x} \ln 2/\sqrt{n}$ exactly. The parametric bootstrap reproduces the exact answer when one exists – and keeps working (for the 10th percentile, a ratio of lifetimes, ...) when it does not.

Nonparametric vs Parametric: Side by Side

	Nonparametric	**Parametric**
Estimate of F	$\hat{F}_n$ (empirical)	$F_{\hat{\theta}}$ (fitted model)
Simulation step	resample observed x_i with replacement	draw fresh values from $F_{\hat{\theta}}$
Main assumption	data are i.i.d. and representative	the model family is correct
Strength	robust to model misspecification	more accurate when model is right; works for small n and tails
Main risk	poor for very small n; cannot go beyond observed range	silently wrong if model is wrong



Nonparametric vs Parametric: Side by Side

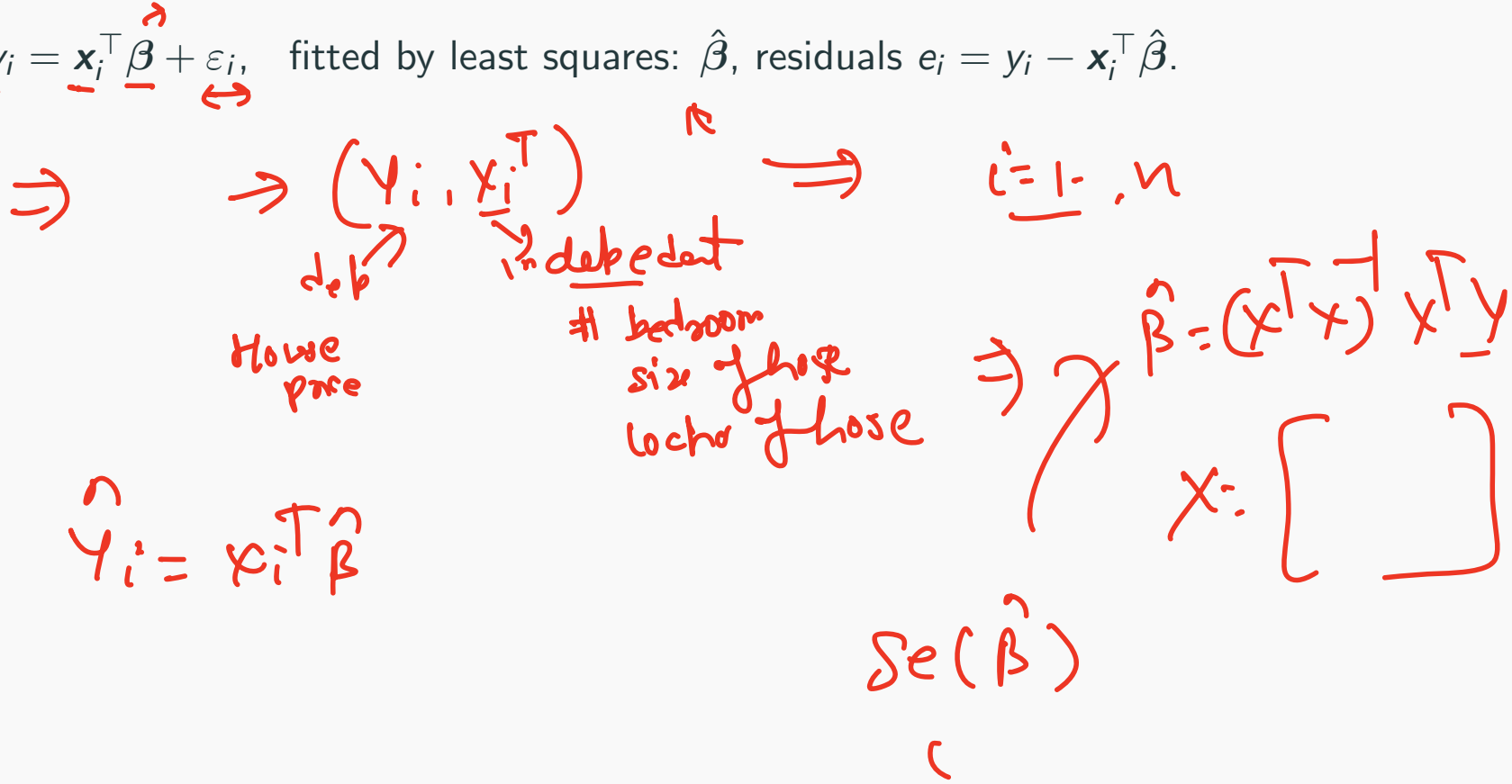
	Nonparametric	**Parametric**
Estimate of F	$\hat{F}_n$ (empirical)	$F_{\hat{\theta}}$ (fitted model)
Simulation step	resample observed x_i with replacement	draw fresh values from $F_{\hat{\theta}}$
Main assumption	data are i.i.d. and representative	the model family is correct
Strength	robust to model misspecification	more accurate when model is right; works for small n and tails
Main risk	poor for very small n; cannot go beyond observed range	silently wrong if model is wrong

A useful habit

Run **both**. If they agree, report with confidence. If they disagree, the disagreement is itself a diagnostic – often of model misspecification.

Bootstrapping Regression Models (in Lab Exercise)

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.



Bootstrapping Regression Models (in Lab Exercise)

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.

Pairs (case) bootstrap

1. Resample pairs $(\boldsymbol{x}_i, y_i)$ with replacement.
2. Refit to get $\hat{\boldsymbol{\beta}}^*$.

+ valid under heteroscedasticity.
– design changes across resamples.

Bootstrapping Regression Models (in Lab Exercise)

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.

Pairs (case) bootstrap

1. Resample pairs $(\boldsymbol{x}_i, y_i)$ with replacement.
2. Refit to get $\hat{\boldsymbol{\beta}}^*$.

+ valid under heteroscedasticity.
– design changes across resamples.

Residual bootstrap

1. Resample residuals $e_1^*, \ldots, e_n^*$ from $\{e_i\}$.
2. Set $y_i^* = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}} + e_i^*$; refit.

+ keeps the design fixed (designed experiments).
– assumes i.i.d. errors.

Bootstrapping Regression Models (in Lab Exercise)

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.

Pairs (case) bootstrap

1. Resample pairs $(\boldsymbol{x}_i, y_i)$ with replacement.
2. Refit to get $\hat{\boldsymbol{\beta}}^*$.

+ valid under heteroscedasticity.
– design changes across resamples.

Residual bootstrap

1. Resample residuals $e_1^*, \ldots, e_n^*$ from $\{e_i\}$.
2. Set $y_i^* = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}} + e_i^*$; refit.

+ keeps the design fixed (designed experiments).
– assumes i.i.d. errors.

- The residual bootstrap is **semi-parametric**: the mean structure is a model, the error distribution is empirical.

Bootstrapping Regression Models (in Lab Exercise)

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.

Pairs (case) bootstrap

1. Resample pairs $(\boldsymbol{x}_i, y_i)$ with replacement.
2. Refit to get $\hat{\boldsymbol{\beta}}^*$.

+ valid under heteroscedasticity.
– design changes across resamples.

Residual bootstrap

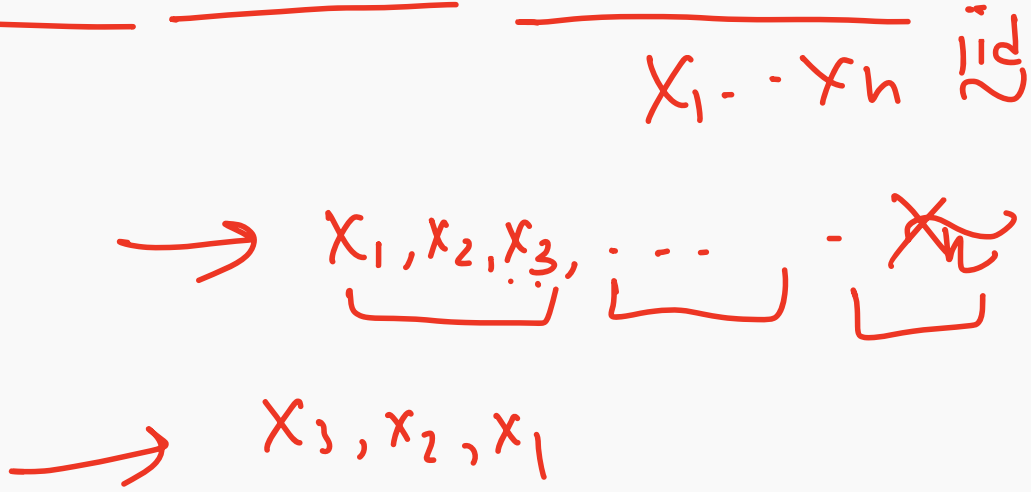
1. Resample residuals $e_1^*, \ldots, e_n^*$ from $\{e_i\}$.
2. Set $y_i^* = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}} + e_i^*$; refit.

+ keeps the design fixed (designed experiments).
– assumes i.i.d. errors.

- The residual bootstrap is **semi-parametric**: the mean structure is a model, the error distribution is empirical.
- Medicine example: dose–response regression from a clinical study – pairs bootstrap is the safe default because variability usually grows with dose.

Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.

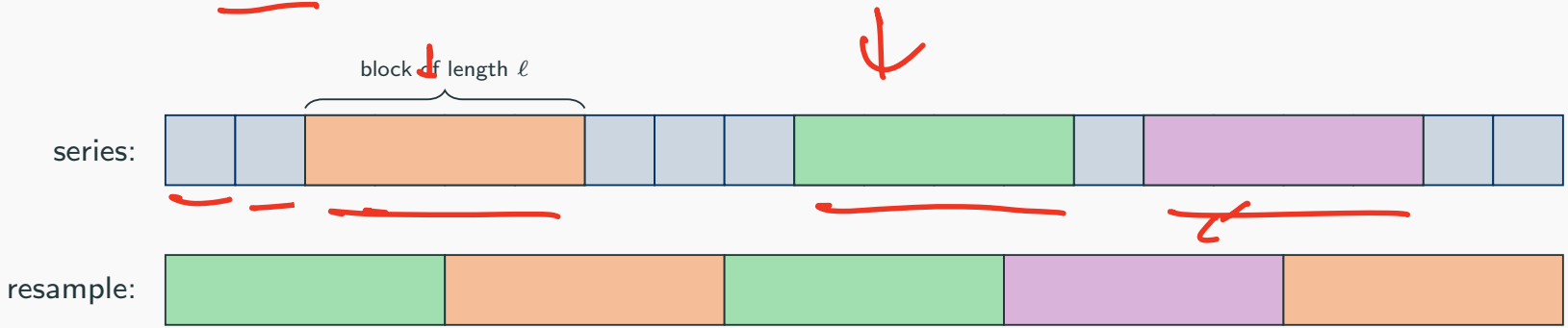


Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.

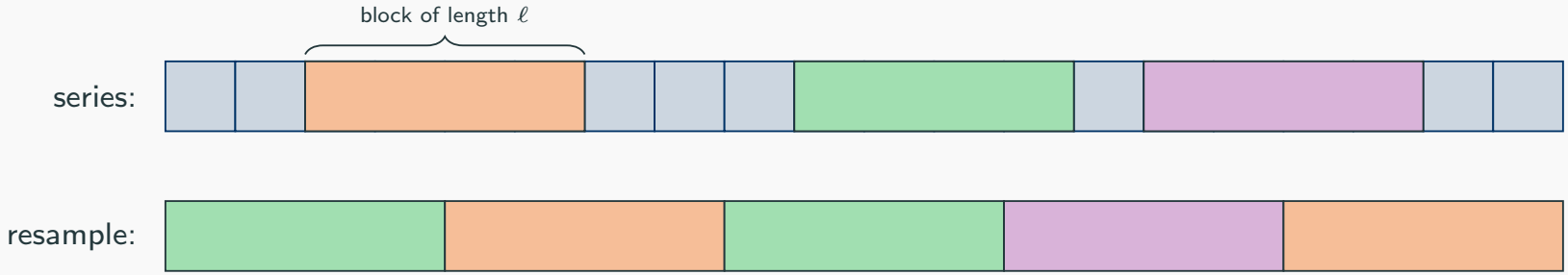
Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.



Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.



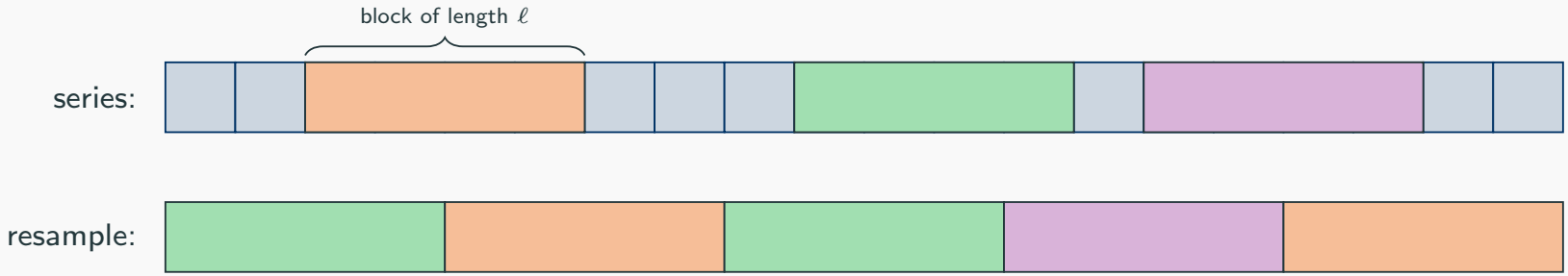
- Within each block the dependence is preserved; it is broken only at the n/ℓ joins.

$X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8$

$X_4, X_5, X_6, X_1, X_2, X_3$

Dependent Data: The Moving Block Bootstrap

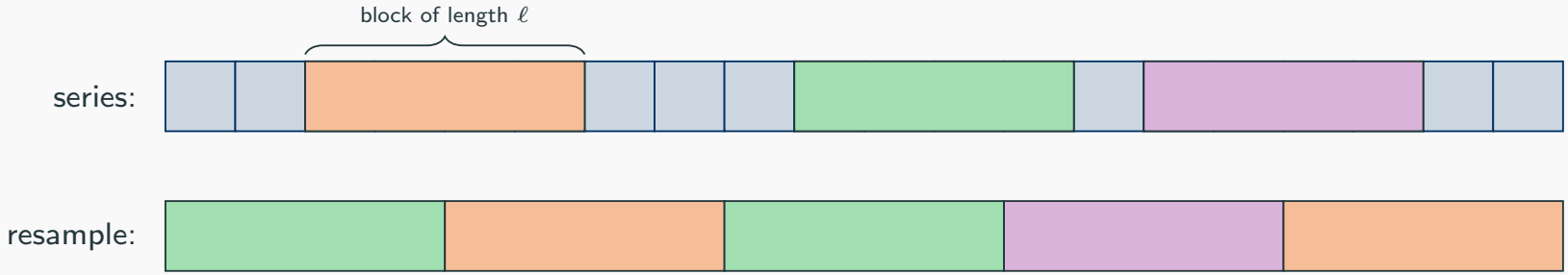
- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.



- Within each block the dependence is preserved; it is broken only at the n/ℓ joins.
- Block length trades bias against variance.

Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.



- Within each block the dependence is preserved; it is broken only at the n/ℓ joins.
- Block length trades bias against variance.
- **Climate:** trends in monsoon rainfall need block (e.g. by season) resampling – an i.i.d. bootstrap gives overconfident intervals.

$X_1, X_2, \ldots X_n$
$\Rightarrow X_1 = X_2 = \cdots = X_n$

Which Bootstrap When?

Data situation	Bootstrap	Resampling unit
i.i.d., no trusted model	Nonparametric	observations x_i
i.i.d., trusted parametric model, small n	Parametric	fresh draws from $F_{\hat{\theta}}$
Regression, random design / heteroscedastic	Pairs	cases $(\boldsymbol{x}_i, y_i)$
Regression, fixed design, i.i.d. errors	Residual	residuals e_i
Time series / spatial data	Block	blocks of neighbours
Clustered data (patients, schools)	Cluster	whole clusters

Which Bootstrap When?

Data situation	Bootstrap	Resampling unit
i.i.d., no trusted model	Nonparametric	observations x_i
i.i.d., trusted parametric model, small n	Parametric	fresh draws from $F_{\hat{\theta}}$
Regression, random design / heteroscedastic	Pairs	cases $(\boldsymbol{x}_i, y_i)$
Regression, fixed design, i.i.d. errors	Residual	residuals e_i
Time series / spatial data	Block	blocks of neighbours
Clustered data (patients, schools)	Cluster	whole clusters

The golden rule

Resample the way the data were generated. The bootstrap replicates whatever randomness you build into the resampling step – and nothing else.